

BHUVISH MAHADEVA

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EDUCATION

Iowa State University

Bachelor of Science | Computer Science and Cybersecurity (GPA: 3.4)

Aug 2023 – May 2027

Ames, IA, USA

PROFESSIONAL EXPERIENCE

Wellmark Blue Cross Blue Shield

Cloud Analyst Intern

May 2026 - Present

Des Moines, IA, USA

- Built a reusable Terraform module for Cisco IOS ACLs, standardizing network security configurations across environments and reducing manual configuration time by 4 hours per deployment
- Developed a Terraform module to automate S3 bucket provisioning across AWS workload accounts, applying Infrastructure as Code practices to cloud resource deployment
- Automated scrubbing of sensitive data from Cisco and Palo Alto Panorama configuration exports prior to GitHub uploads, reducing manual review effort and improving configuration security
- Automated Cisco IOS device configuration uploads to GitHub, enabling AI-assisted analysis of network device configuration files and improving version control and configuration management workflows

Shadowfax

Cyber Security Analyst Intern

May 2025 – Aug 2025

Bengaluru, KA, IND

- Developed and deployed Python-based phishing and threat-detection tools in cloud-hosted environments, automating IOC extraction and reducing incident response time by 40%, processing 200+ alerts monthly
- Integrated threat intelligence across MISP, PhishTool, Wazuh, Zeek, mirroring AWS multi-service troubleshooting by Correlating Logs, Alerts, and Network Telemetry across multiple platforms
- Documented Incident response workflows aligned with ISO 27001, applying AWS-style support practices including RCA creation, Ticket lifecycle management, and Customer-focused communication
- Delivered monthly phishing simulations for 250+ employees, demonstrating customer-support readiness and improving organizational resilience with a 25% reduction in successful attacks and 60% increase in reporting

Iowa State University

Software Engineer

May 2024 – Aug 2024

Ames, IA, USA

- Engineered backend systems for Eaton fire panel servers using cloud-aligned architectural principles, improving real-time alert reliability through patterns like AWS CloudWatch and event-driven monitoring
- Analyzed fire safety data with Python, applying troubleshooting workflows common to AWS Support, including log inspection, metric correlation, and anomaly detection
- Designed and optimized SQL databases, reducing data retrieval time by 25%, demonstrating database fundamentals transferable to AWS RDS and DynamoDB performance tuning
- Built and integrated BIM (Building Information Modeling) features using Java/JavaFX for real-time visualization, cutting simulation time by 30% and improving cross-team operational visibility

RELEVANT PROJECTS

Honeypot & SIEM: Real-World Attack Detection on AWS

Mar 2026 – Present

- Built a full attack-detection pipeline on AWS, deploying Cowrie SSH honeypot and OpenCanary to lure attackers, with logs shipped via Filebeat to Elasticsearch and visualized in a Grafana dashboard.
- Captured 737 real-world attacker events over 4 days, including a single IP responsible for 477 brute-force attempts and a Microsoft Azure-based attacker.
- Mapped key findings to MITRE ATT&CK techniques (T1110 Brute Force, T1046 Network Scanning, T1078 Valid Accounts) to contextualize detection results.

SIEM Log Analyzer

Mar 2026 – Present

- Built a full-stack Security Information & Event Management (SIEM) system from scratch, including a real-time log ingestion engine supporting Syslog, Apache/Nginx, JSON, and Windows Event Log formats.
- Implemented a rule-based threat detection engine with 9 built-in detection rules covering SSH brute force, SQL injection, malware execution, and data exfiltration.
- Built anomaly detection to automatically flag IP burst activity exceeding 10 events per 30 seconds, with full incident management including MITRE ATT&CK mapping and status lifecycle tracking.
- Integrated WebSockets for live alert streaming to a React dashboard.

LabFlow

Jan 2026 – May 2026

- Built backend services in Java for LabFlow, an app that parses lab manuals and auto-fills student answers back into documents, helping students complete labs up to 60% faster.
- Designed and managed the SQL database schema to store lab manuals, parsed steps, and student responses, ensuring efficient querying and data integrity.
- Implemented backend logic to break down lengthy lab documents (up to 50 pages) into structured, digestible steps.
- Collaborated with a 4-person team to build and ship LabFlow within a semester, winning 2nd Place Overall and 1st Place in-section at COM S 3090 (Spring 2026).

Hospital Network Penetration Test Lab

Nov 2025 – Dec 2025

- Conducted end-to-end penetration testing using Nmap, Nikto, Metasploit to exploit MS17-010, anonymous FTP, weak SSH, and credential misconfigurations.
- Delivered a non-technical executive report and technical documentation including root-cause findings, PHI exposure risks, and detailed remediation steps.

Automated Brand Threat Detection & Takedown System

May 2025 – Aug 2025

- Built an automated Python-based system for detecting homoglyph, typo-squatted, and phishing domains integrating WHOIS, DNS, AbuseIPDB, and social media monitoring
- Flagged 200+ malicious domains, reducing manual security monitoring time by 70%
- Implemented logging pipelines, scheduled alerting, and takedown request workflows

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, S3, IAM, CloudWatch, Lambda fundamentals, CloudFormation, Ansible), Terraform, Azure, GCP, virtualisation (Proxmox, VMware), Linux administration (Ubuntu, CentOS).

Programming: Python, Java, C/C++, JavaScript, React.js, HTML/CSS.

Networking: DNS, TCP/IP, HTTP, SSH, packet analysis (Wireshark, Zeek), Palo Alto Panorama, Cisco IOS ACLs, VLANs, Logging, AAA Authentication.

Databases: SQL, relational database design, data retrieval optimisation.

Security & Tools: Nessus, MISP, Wazuh, Splunk, Kali Linux, Metasploit, SIEM workflows, threat detection automation, incident response.

Software Engineering: Backend development, API integration, DevOps basics, CI/CD exposure, Git workflows.

Troubleshooting: Root-cause analysis, log inspection, multi-service dependency debugging, and customer issue resolution.